

Comparing Vulnerabilities in Networks: Software Defined Networks in Personal Applications

Maxwell Stratford

May 7, 2026

1 Introduction

With the ever-changing landscape of the Information Technology (IT) field, it is important to stay up to date on security vulnerabilities and new technologies that are being developed. However, someone who does not work with technology will not be aware of these vulnerabilities or concepts in network security, even though their networks may be the easiest to attack. According to a survey done by the Broadband Genie company in 2025 [2], 47% not adjusted their routers factory settings, 81% the WIFI password, and 79% These vulnerabilities that can be easily mitigated, but are not, add to the problem of home internet security. For the people out there, who need something else to grasp the seriousness of internet security, the FBI put out a “Internet Crime Report” [?] where they mention the 16 billion reported dollars stolen through cyber-crimes in 2024. One of the top ways mentioned in the report for a threat actor (hacker) to do this is through a man in the middle (MITM) attack, which will be mentioned and mitigated in this paper. With the right setup and teaching however, these personal networks can be easily secured. One of the newer technologies in the field that can help towards securing these networks is a Software Defined Network (SDN) controller. These devices allow for a more flexible, efficient, and secure network by separating the control plane which controls decision making, from the Data Plane which controls the portion of the network that handles the actual data transfer itself. By centralizing the decision making to the SDN controller, it abstracts the underlying hardware and allows for more efficient network management. The goal of this paper is to compare a basic network setup in a personal network falling to a MITM attack, then to a setup with a SDN controller showing the attack failing with the SDN implementation. This comparison is important because it will help to inform people about security vulnerabilities, give them tools to mitigate these vulnerabilities, and create a more secure network. Prior works to this paper, including [1], dive into enterprise applications for this technology, and how it can be used to mitigate large scale attacks. However, this paper will focus on the personal networks that are often overlooked. Towards this goal, VirtualBox will be used to simulate a network and its traffic, while Kali Linux will be used to penetration test the network. The attack that will be performed during the penetration testing is MITM. The hope for this paper is to alert the

average user to security vulnerabilities in their network and give them the tools they need to mitigate them. One such as a SDN controller, which this paper evaluates through its simulated attacks.

2 Background

This section will cover the background information that is necessary to understand the rest of the paper, as well as a home network setup. This includes the different types of networking devices and software tools.

2.1 How Networking Devices Work Together

An end device can be classified as a source or destination for internet traffic or data packets. This device can be anything ranging from a laptop or phone, TV or printer, or even an internet of things (IoT) device such as a home thermostat. An end user is using a laptop to connect to a website. It does this through sending a request, on most home networks, to the router. The router will perform a network address translation (NAT), to change the private IP address of the laptop, into an internet protocol version four (IPv4) address of its own public router IP address, so the request can get to the internet. These addresses tend to be formatted as such. 192.168.1.X. It sends a request to a internet service provider (ISP) for the contents of the IPv4 address provided, which is then replied to with the contents of the website. The router will then send the laptop the requested information to load the webpage. That is how a traditional network would work. With a software defined network controller (SDN), things change slightly. An SDN controller works alongside switches in a network. A switch is placed between the end devices and the router to handle traffic within the network to take a load off the router. The SDN controller sits outside of the data path, which allows it to program the switch with forwarding rules in place using different protocols. When data hits this switch, the switch follows the controllers' rules without querying the controller each time, unless a rule does not exist. In short, a traditional network distributes its decision making and forces the router to make more decisions, while an SDN centralizes all of these decisions in its controller and lets other devices just forward the data. This separation of the control plane and the data plane is what enables stronger security policies and allows mitigation of different network attacks like MITM.

2.2 Traditional Network Devices

The router is the last point in a network that data will pass through. Its main purpose it to manage the network and allow local users to connect to the internet through the domain name service (DNS) which converts URLs to IPv4 addresses for data transfer, and NAT for converting a local IP address into its own to allow the ISP to send the requested data back to the router. It also, in a basic network, assigns IP addresses to local users, This is usually a bigger box with multiple cables coming out of it provided by an ISP. Some security concerns which were brought up in the introduction include statistics like, 79% of people do not know how to change their default settings, so anyone who can get onto this network,

can then get into the router settings and control everything about the network. A common attack type against home routers is called a Brute-Force attack against the admin login credentials. Because of 81% of users never changing the default username and password to their router settings, an attacker on the local network can try pairs like admin/admin or root/root until they gain access. The switch is a basic device. Their main goal is to forward data between the router and other end devices. Most of these look like a small box with two to six ports on the front. With one port having an internet symbol to connect to the router, and the other points used for the end devices. These devices make the most basic forwarding decisions, passing all data based on a forwarding table they create from the data they pass on. Switches are vulnerable to ARP spoofing because they trust all ARP replies that they receive. An attacker sending forged packets back can trick the switch into sending traffic meant for the router to the attackers machine instead. End devices are any device that sends or receives data on the network, Laptops, phones, printers, IoT devices, cameras, ect. They can be vulnerable on the network if not updated or managed correctly. Attackers can target these devices more easily if updates are not installed regularly, and most people at least know about attacks aimed at these devices like spam emails or texts. End devices are often the weakest link in any network, home or enterprise. Users can get behind on updates, and many devices never get security patches after the point of sale. The SDN controller is a device outside of the normal line of local network traffic that oversees managing the switches. This changes the switches use in the network by taking control away from the switch entirely, giving all power to the controller. It would physically be near the switch and be a raspberry Pi or an old PC. Most people will use OpenFlow for this service and run it on either of the two devices mentioned. It improves security by allowing the controller to see the entire network and will enforce global (network wide) policies that block attacks like ARP spoofing to occur. It does this by separating the control plane (decision-making) and data plane (packet forwarding). The SDN controllers centralized view of the entire network allows it to detect anomalies that a traditional router would end up missing. For example, if the same MAC address claims to have multiple IP addresses, the controller can flag this and block any further traffic from that device by blocking the port.

2.3 MITM Attack

During a man in the middle attack, an attacker secretly intercepts data being transferred to another source, and potentially alter the communication between the two parties, who will still believe they are in communication with one another. This paper will use the technique called address resolution protocol (ARP) poisoning or ARP spoofing. ARP usually works by mapping Ip addresses to their perspective MAC addresses inside of a local network. Normally, a device would ping another and ask, in human terms, “who has the IP address of 192.168.1.1”. The router in most cases would be the one to respond to that ping in a network with default settings. ARP spoofing works by forging replies. A threat actor will claim that their MAC address maps to that IP address so the victims ARP table will update to send all data to the attacker. Once an attacker is in the middle of this communication, they have a few choices. They can passively listen to get data without interacting with it or they can actively modify the data in transit to redirect to fake website or inject malicious code. The scripts that were used in the Black Hat Python book worked by enabling IP

forwarding so that the victim's traffic still reaches the router, sends forged ARP packets, and uses a packet sniffer to capture the traffic. The standard home network is vulnerable to this because there are no checks on active ARP tables to make sure everything looks correct, local traffic is often unencrypted, and the users on these machines have no easy way to detect ARP poisoning.

2.4 Other Attack Types

While this paper focuses on MITM attack using ARP Spoofing or Poisoning, home networks often face a variety of other threats. Understanding these attacks helps show why SDN controllers are so valuable, even beyond just stopping ARP attacks. DNS Spoofing (DNS Cache Poisoning) targets the domain name resolution process. When a user types a website address like `www.wooster.edu`, their device sends a DNS request to a DNS server. In a DNS spoofing attack, the attacker injects a fake DNS response to a user, redirecting their traffic to a malicious website even though they typed in the correct address. The fake website looks completely identical to the correct one, tricking users into inputting all of their account details and information or downloading a piece of malware. Traditional home routers have no protection against this sort of DNS spoofing because they simply forward DNS traffic onward without inspection. Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks aim to overwhelm a network device with so much traffic that no data can leave the device. In a home network, an attacker would floor the router with so many requests, junk data, or packets that it becomes unresponsive, severing the local network from the internet. While home users are less likely to be targeted by a larger scale DoS attack, the DDoS, DoS attacks, can still occur from a single compromised device on the local network. An SDN controller could help limit the rate of traffic going to the router or drop malicious packets in their entirety. The Evil Twin Attack involves creating a fake Wi-Fi access point with a real sounding name, such as `Free ____ Wi-Fi` or `Home Network 2.4GHz`. When nearby devices connect to this fake network, the attacker can capture all of the data passing through the network. This includes passwords, raw data and other sensitive information. This attack is especially dangerous in public places but can also be used against a home network if the attacker is physically nearby. Because the fake network can look the same as compared to the real one, users would have no easy way to know if they are being attacked. Brute Force and Password Attacks target weak router admin usernames and passwords. As noted in the introduction, 81% of users never change their router's default credentials. Attackers can use automated tools to try all of the common username and password combinations until they gain access to a network. Once inside the router, an attack can change the DNS settings, disable other security features, and/or forward all traffic to their own computers. This attack is easy to execute and remains effective because most home users do not think to change the default credentials.

3 Related Works

Biswas [1] investigates SDN-based mitigation of different network attacks such as distributed denial of service (DDoS). By comparing an SDN setup to a traditional network, Biswas

demonstrated that the SDN improved both detection and response times. However, this study focuses solely on enterprises, or big business, networks with dedicated IT teams and high-end hardware. These findings do not show how it could effect a home network setup, where users have less technical knowledge and routers are often left with default settings. Rostami et al. [5] examined internet facing router services and found that over 40% of known routers IPv4 addresses had at least one vulnerable service exposed. The failure to follow basic security practices, especially in these common home networks where most users do not have a technical knowledge to secure their devices or may not even know it is necessary leave them open to attack. As noted in the introduction, surveys show that 81% of users never change their routers admin password, and 79% would not know how to if needed. Network simulation tools like VirtualBox [4] and Cisco’s Packet Tracer [3] are widely used to model network behaviors and test security configurations. However, existing studies use these tools to simulate large business environments (enterprise networks). There is a noticeable gap in works done regarding simulated home networks, where device types, traffic patterns, and user behaviors are significantly different. This paper aims to fill that gap by simulating a home network on VirtualBox, then implementing an SDN controller. Unlike prior works that focus on DDoS mitigation in an enterprise setting, this study will focus on MITM attacks in a home network, which is a scenario which is far more likely for a home user.

4 Methodology

To compare SDNs in enterprise and personal networks, this study uses a simulation-based approach. Through simulating such networks with their traffic, and then performing penetration testing on them, we can see the vulnerabilities of each network, and how the SDN can mitigate those vulnerabilities.

4.1 The Simulation Environment

The simulation environment was created using Oracle’s VirtualBox. Four virtual machines were used in this process. A Windows 10 machine acting as the victim, a Kali Linux machine acting as the attacker and hosting the SDN controllers’ software, a router VM configured with default settings, and a switch to be managed during the SDN mitigation. All VMs were connected through a virtual, virtual local area network (VLAN) to isolate the simulation from the host machine and the internet. The Windows machine had a sample file downloaded to it for the attacker to target and the Kali Linux machine was preloaded with network penetration testing tools. To execute the MITM attack on the home network without an SDN, the attacker first loaded a listener.py file to the windows Machine. A proxy and a Socket were ran in order to be able to communicate with that device. Any command ran on the socket would then show the output on the Kali Linux machine. Through this method, you can get any information off the Windows machine that can be gathered from a command line input. Shown in Figure 1, the attacker successfully captures directory listings, application names, and raw hexadecimal data from the targeted computer. The script was adapted from the MITM chapter in Black Hat Python [6], which provides code for ARP spoofing and packet sniffing. After the initial attack succeeded, an SDN controller was implemented. To

save resources on the host laptop for this project, the SDN controller was actually hosted on the Kali Linux machine. The controller communicated with an Open vSwitch (OVS) placed between the end devices and the router. The controller was programmed with a security policy that inspected ARP traffic and allowed only the router’s MAC address to reply to ARP requests for the default gateway. When the attacker attempted to forge replies, the packets were dropped. With the SDN in place, no data was captured. This is shown in Figure 2, where the code had to be interrupted to stop running after five minutes. In summary, the same MITM attack succeeded on the traditional home network, capturing sensitive data about the user’s Windows machine. After implementing the SDN controller with a simple policy, the same attack was completely mitigated. This comparison shows how effective the use of SDN controllers can be in a home network environment.

```

Session Actions Edit View Help
[venv2]-(kali@kali)-[~/bhp]
└─$ python proxy.py 9999 192.168.1.100 7777
[*] Listening on 0.0.0.0:9999
[*] Got connection from 127.0.0.1:58186
[*] Connected to 192.168.1.100:7777

[==>] Received 4 bytes from client
0000 04 09 72 0a dir.

[==>] Received 526 bytes from remote
0000 20 56 6f 6c 75 6d 65 20 69 6e 20 64 72 69 76 65 Volume in drive
0010 20 43 20 69 73 20 57 69 6e 64 6f 77 73 0d 0a 20 C is Windows..
0020 56 6f 6c 75 6d 65 20 53 65 72 69 61 6c 20 4e 75 Volume Serial Nu
0030 6d 62 65 72 20 69 73 20 33 32 39 32 20 41 32 39 mber is 3292-A29
0040 30 0d 0a 0d 0a 20 44 69 72 65 63 7a 6f 72 79 20 0.... Directory
0050 6f 66 20 43 3a 5c 55 73 65 72 73 5c 55 73 65 72 of C:\Users\User
0060 5c 44 65 73 6b 7a 6f 70 0d 0a 0d 0a 30 34 2f 32 \Desktop....04/2
0070 29 2f 32 30 32 3e 20 20 20 34 2a 20 35 20 50 4d 9/2026 04:05 PM
0080 20 20 20 20 3c 44 49 52 3e 20 20 20 20 20 20 <DIR>
0090 20 20 20 2e 0d 0a 30 34 2f 32 39 2f 32 30 32 36 ... 04/29/2026
00a0 20 20 3a 3a 30 3a 20 50 4d 20 20 20 20 3c 44 04:04 PM <D
00b0 49 52 3e 20 20 20 20 20 20 20 20 20 20 2e 0d IR> ...
00c0 0a 30 33 2f 30 39 2f 32 30 32 36 20 20 30 37 3a .03/09/2026 07:
00d0 35 35 20 41 4d 20 20 20 20 20 20 20 20 20 20 55 AM
00e0 33 39 32 2c 33 32 30 20 46 69 72 65 66 6f 78 2e 392,320 Firefox.
00f0 65 78 65 0d 0a 30 33 2f 31 31 2f 32 30 32 36 20 exe..03/11/2026
0100 20 30 35 3a 31 38 20 50 4d 20 20 20 20 20 20 05:18 PM
0110 20 20 20 20 20 32 2c 33 35 39 20 4d 69 63 72 2,359 Micr
0120 6f 73 6f 66 7a 20 45 64 67 65 2e 6c 6e 6b 0d 0a osoft Edge.lnk..
0130 30 34 2f 32 39 2f 32 30 32 36 20 20 30 34 3a 30 04/29/2026 04:0
0140 3a 20 50 4d 20 20 20 20 20 20 20 20 20 20 20 4 PM
0150 20 31 2c 3a 30 30 20 50 69 73 61 6c 20 53 7a 1,408 Visual St
0160 75 64 69 6f 20 43 6f 64 65 2e 6c 6e 6b 0d 0a 30 udio Code.lnk..0
0170 3a 2f 32 39 2f 32 30 32 36 20 20 30 35 3a 32 37 4/29/2026 05:27
0180 50 4d 20 20 20 20 20 20 20 20 20 20 20 20 20 PM
0190 20 20 33 38 32 20 77 69 6e 64 6f 77 73 5f 6c 69 382 windows_li

```

Figure 1: This is a screenshot of the data that was captured on the Windows machine during the MITM attack. It was captured after running two commands, one to get the windows machine ready, and the other to pull data onto the Linux Machine. The data that was captured was the output of the directory command, which shows the volume for the drive, the serial number, applications installed, and the raw hexadecimal data.

5 Conclusion

Home networks remain dangerously vulnerable to attack, yet they receive far less attention in security research than enterprise networks. As noted in the introduction, 81% of users never change their router’s default admin username and password, and 79% would not know how to change any settings if needed. This paper addressed this gap by simulating a typical home network using VirtualBox and comparing network security in a network without an SDN and with an SDN. A man in the middle (MITM) attack using ARP spoofing was executed against both configurations. The MITM attack succeeded on the average home network, capturing directory listing, application names, and raw hexadecimal data from the Windows 10 victim machine. As shown in Figure 1. When the same attack was attempted after implementing an SDN controller, the attack was completely mitigated. The controller,

```
(kali@kali)-[~]
$ cd ~/bhp
source venv3/bin/activate
python proxy.py 9000 192.168.1.102 7777
[*] Listening on 0.0.0.0:9000
whoami
^CTraceback (most recent call last):
  File "/home/kali/bhp/proxy.py", line 80, in <module>
    server_loop(local_port, remote_host, remote_port)
    ~~~~~^~~~~~
  File "/home/kali/bhp/proxy.py", line 62, in server_loop
    client_socket, addr = server.accept()
    ~~~~~^~~~~~
  File "/usr/lib/python3.13/socket.py", line 295, in accept
    fd, addr = self._accept()
    ~~~~~^~~~~~
KeyboardInterrupt
```

Figure 2: This figure shows the output of the same attack type, same code, same virtual machines, but with the implementation of an SDN in the network. As can be seen, the attack was not successful, and the attacker was not able to capture any data from the Windows machine. This further shows the effectiveness of the SDN in mitigating vulnerabilities in the home network.

programmed with a simple policy, allowed only the routers MAC address to reply to ARP requests, mitigating the ARP poisoning attack before packets could reach the victim. Figure 2 shows how the attack had to be stopped after five minutes due to getting nothing from the network. This demonstrated the separation between the control plane and the data plane in the network, which is a core feature of an SDN, enabling security policies that are impossible in the traditional home network setup. This study however did have its limitations. The network was simulated entirely in VirtualBox and was not deployed on physical hardware with real internet traffic. Additionally, only one type of attack was tested (ARP Spoofing). In future work, examining how an SDN controller can mitigate other types of attacks such as DNS spoofing or DDos attacks could be beneficial in understanding how they could make a home network more secure. This could all be done in an existing home network with the addition of a Raspberry Pi. Despite these limitations, this paper has clearly demonstrated the effectiveness of the SDN in mitigating MITM attacks in the home network setting. For the average user, the first step should be changing the default router username and password, the next best step is setting up an SDN to mitigate attacks that have made it into the network. For those who are looking to do so, a Raspberry Pi running OpenVSwitch could be a cheap and easy way to start securing their own network. The implications of this study could extend beyond the specific MITM attack tested. If the SDN controller can become a more affordable and user-friendly device, they could become as common as home routers are today. A plug-and-play SDN device could be implemented in a next gen router and then could automatically enforce security policies on a home network that currently would require a lot of configuration. While this study used a simulation, the underlying principles apply to real world hardware. The same Open vSwitch and controller software tested here can run on a Raspberry Pi device costing less than one hundred dollars, making security accessible to home users ready to implement real security in their networks.

6 Addendum: The Use of Artificial Intelligence

Generative AI tools were used in a limited and supporting way throughout the writing of this paper. Their use was restricted to brainstorming, clarification of technical concepts, finding sources, and formatting of the Methodology section. The actual MITM attack script used in this study was adapted directly from the Black Hat Python. When looking at difficult code examples, ChatGPT was occasionally used to describe what the code was doing in the given context, closely relating to how one would use the website Stack Overflow. Most prompts were used to ask technical questions relating to error messages from a Windows terminal. At no point was AI used in any way that would be against the Wooster Ethic.

References

- [1] BISWAS, R. *Mitigation of Different Network Attacks and Optimization in Software Defined Network*. Ph.d. dissertation, ProQuest Dissertations Publishing, 2021. ProQuest Dissertations & Theses A&I.
- [2] BROADBAND GENIE. Router security survey 2025. Broadband.co.uk, Apr. 2026. Survey of 3,242 users; 47% never adjusted router factory settings, 81% never changed admin password.
- [3] CISCO SYSTEMS, INC. Cisco packet tracer download and installation instructions. Online PDF. Accessed: 2026-04-08.
- [4] ORACLE CORPORATION. Virtualbox documentation. Online. Accessed: 2026-04-08.
- [5] ROSTAMI, S., HEINRICH, T., AND ALBAKOUR, T. Poster: An investigation into internet-facing router services. In *Proceedings of the 2024 ACM on Internet Measurement Conference* (New York, NY, USA, 2024), IMC '24, Association for Computing Machinery, p. 775–776.
- [6] SEITZ, J., AND ARNOLD, T. *Black Hat Python: Python Programming for Hackers and Pentesters*, 2 ed. No Starch Press, 2021.